

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: MLA03

COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO3: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).*

Assessment Tool Weightage: 25%

Dr.T.P. Anithaashri

QUESTIONS: 10

Total Marks: 50

Annexure A

Data Interpretation

Duration: 1hr

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.

(5 Marks)

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance.

(5 Marks)

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results.

(5 Marks)

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance. **(5 Marks)**

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data. **(5 Marks)**

6. An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability. **(5 Marks)**

7. A Q-learning agent is tested with different exploration strategies: ϵ -greedy, softmax, and random exploration. The cumulative reward is highest with ϵ -greedy. Compare the strategies and defend why ϵ -greedy achieves better performance in this scenario. **(5 Marks)**

8. An RL agent trained in a maze environment initially takes 50 steps to reach the goal, but after 300 episodes it reduces to 15 steps. Interpret the reduction in steps and evaluate whether this indicates convergence to an optimal policy. **(5 Marks)**

9. A SARSA agent is trained with discount factors of 0.4, 0.7, and 0.95. The agent achieves the highest long-term reward with 0.95. Discuss the role of discount factor in balancing immediate vs. future rewards and justify why 0.95 is most effective. **(5 Marks)**

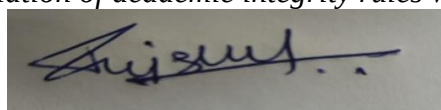
10. An RL agent trained in a stochastic environment shows fluctuating rewards in early episodes but stabilizes after 500 episodes. Examine the reward trend and defend how stability indicates policy improvement and robustness of learning. **(5 Marks)**

Code of Conduct:

I B. Rajesh (192472165) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

signature of the student:



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

ASSESSMENT TOOL-3

B. Rajesh

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.

Answer:

The reward obtained by an RL agent is an important measure of its performance. In the given case, the reward gradually increases from 10 to 45 during training.

At the beginning, the agent has very little information about the environment. It tries different actions and may make several wrong decisions. Because of this, the initial reward is low.

As more episodes are completed, the agent learns from the results of its previous actions. It identifies useful actions and avoids actions that lead to poor results. This causes the reward to increase gradually.

The increasing reward also shows that the learned policy is becoming better. If the reward continues to remain high and stable after further training, it indicates that the agent has learned a reliable strategy and may have reached convergence.

Conclusion:

The increase from 10 to 45 indicates successful learning. The agent is making better decisions, improving its policy, and moving toward a stable and effective solution.

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance.

Answer:

The epsilon-greedy method is used to control the balance between exploration and exploitation.

Exploration means trying new actions to find better possibilities. Exploitation means using the action that the agent currently believes is the best.

When $\epsilon = 0.1$, the agent explores only occasionally and mostly selects the best-known action. Therefore, it spends more time making useful decisions and obtains the highest cumulative reward.

When $\varepsilon = 0.3$, the agent performs more exploration. This can help it discover new actions, but some of those actions may produce lower rewards.

When $\varepsilon = 0.5$, the agent explores very frequently. Half of its decisions may be exploratory, so it may ignore good actions that it has already learned. This explains why the cumulative reward is lowest.

Conclusion:

In this situation, $\varepsilon = 0.1$ is the best value because the agent makes greater use of its learned knowledge while still performing some exploration.

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results.

Answer:

Q-Learning and SARSA are two important Reinforcement Learning algorithms used to learn the value of actions.

Q-Learning is an off-policy method. It updates its knowledge using the best possible action that can be taken in the next state.

SARSA is an on-policy method. It updates its value based on the action that the current policy actually chooses.

In the given comparison, Q-Learning produces slightly higher rewards at 100, 200, and 300 episodes. This means that Q-Learning is performing better throughout the training period.

The consistent advantage in reward suggests that Q-Learning is learning an effective policy earlier than SARSA. Therefore, based on the given observations, Q-Learning can be considered to have faster convergence.

Conclusion:

Q-Learning performs slightly better than SARSA in this experiment. The higher rewards across the episodes indicate that Q-Learning reaches a good policy faster.

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and

justify which discount factor is most suitable for achieving better learning performance.

Answer:

The discount factor (γ) determines the importance given to future rewards by the RL agent.

A smaller value of gamma makes the agent concentrate mainly on rewards that can be obtained immediately. A larger value makes the agent consider rewards that may be received several steps later.

For $\gamma = 0.2$, the agent strongly prefers immediate rewards.

For $\gamma = 0.5$, the agent gives a balanced importance to immediate and future rewards.

For $\gamma = 0.9$, the agent gives greater importance to future rewards and can plan its actions for long-term benefits.

Since the experiment shows that the average reward increases when gamma increases, the agent benefits from looking further into the future.

Conclusion:

Among the given values, $\gamma = 0.9$ is the most suitable because it provides the highest reward and helps the agent make better long-term decisions.

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data.

Answer:

The success rate indicates how frequently the agent is able to complete the task correctly.

At 50 episodes, the agent has a success rate of only 45%. This shows that the agent is still in the early learning stage and does not yet have enough experience to make consistently good decisions.

After training for 200 episodes, the success rate increases to 92%. This large increase indicates that the agent has learned from its previous experiences and has improved its action selection.

A success rate of 92% means that the agent is successful in most attempts. Therefore, the learned policy is clearly much better than the policy at the beginning.

However, it is not sufficient to say that the policy is definitely optimal only from this information. More training results would be required to determine whether the success rate has reached a stable maximum.

Conclusion:

The increase from 45% to 92% demonstrates significant learning and policy improvement. The agent has learned a very effective policy and is likely approaching an optimal solution.

6. An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability.

Answer:

The learning rate determines the size of the changes made to the agent's knowledge during training.

With a learning rate of 0.01, the agent makes very small updates. This generally makes learning more gradual and can require more episodes to reach a good solution.

With 0.05, the agent makes moderate updates. This allows it to learn at a reasonable speed without making the learning process unstable.

With 0.1, the updates are much larger. The agent may learn quickly, but large updates can cause the learned values to change too much, resulting in unstable training.

Since the highest stability is observed at 0.05, this value provides a suitable balance between learning speed and stability.

Conclusion:

0.05 is the best learning rate among the given choices because it provides stable learning while still allowing the agent to converge at a reasonable speed.

7. A Q-learning agent is tested with different exploration strategies: ϵ -greedy, softmax, and random exploration. The cumulative reward is highest with ϵ -greedy. Compare the strategies and defend why ϵ -greedy achieves better performance in this scenario.

Answer:

Exploration strategies help the agent decide whether to use known actions or try new actions.

In ϵ -greedy exploration, the agent normally selects the action with the highest estimated value. Occasionally, it selects another action to explore the environment.

In softmax exploration, actions are selected according to their estimated values. Actions with better values receive a higher probability of being selected, but other actions can also be chosen.

In random exploration, the agent chooses actions without giving much importance to their previously learned values.

In this experiment, ϵ -greedy gives the highest cumulative reward because it mainly uses the best-known actions while still providing opportunities for exploration.

Random exploration may result in many poor actions, while softmax may continue selecting lower-valued actions with some probability.

Conclusion:

The ϵ -greedy strategy performs best in this case because it provides an effective balance between exploring new actions and exploiting actions that are already known to be useful.

8. An RL agent trained in a maze environment initially takes 50 steps to reach the goal, but after 300 episodes it reduces to 15 steps. Interpret the reduction in steps and evaluate whether this indicates convergence to an optimal policy.

Answer:

The number of steps required to reach the goal is a useful measure of efficiency in a maze. Generally, a smaller number of steps means that the agent has found a better route.

Initially, the agent requires 50 steps because it is unfamiliar with the maze. It explores different paths and may take unnecessary movements.

After 300 episodes, the number of steps decreases to 15. This indicates that the agent has learned from repeated interactions and discovered a much shorter path.

The reduction in steps demonstrates clear policy improvement. The agent is now selecting actions that take it toward the goal more efficiently.

To confirm that the policy is optimal, we need to know whether 15 is the shortest possible path and whether the agent continues to achieve this result in later episodes.

Conclusion:

The reduction from 50 to 15 steps is strong evidence of learning and improved efficiency. It suggests that the agent is approaching an optimal policy, especially if its performance remains stable.

9. A SARSA agent is trained with discount factors of 0.4, 0.7, and 0.95. The agent achieves the highest long-term reward with 0.95. Discuss the role of discount factor in balancing immediate vs. future rewards and justify why 0.95 is most effective.

Answer:

The discount factor γ controls how strongly the agent considers future rewards when making decisions.

With $\gamma = 0.4$, the agent gives greater importance to rewards that are received immediately and less importance to rewards received later.

With $\gamma = 0.7$, future rewards receive a moderate level of importance.

With $\gamma = 0.95$, the agent gives very high importance to future rewards. This allows it to make decisions that may not provide an immediate benefit but can result in better rewards in later states.

Since the highest long-term reward is obtained with $\gamma = 0.95$, the experiment indicates that future rewards are important for achieving good performance in this environment.

Conclusion:

$\gamma = 0.95$ is the most effective value because it encourages long-term planning and produces the highest overall reward among the tested values.

10. An RL agent trained in a stochastic environment shows fluctuating rewards in early episodes but stabilizes after 500 episodes. Examine the reward trend and defend how stability indicates policy improvement and robustness of learning.

Answer:

A stochastic environment contains some degree of randomness. Therefore, the same action may not always produce exactly the same result.

During the initial episodes, the agent is still learning and exploring different actions. Because of this exploration and environmental randomness, the reward may change significantly from one episode to another.

After around 500 episodes, the rewards become more stable. This indicates that the agent has gained sufficient experience and has learned which actions generally produce better results.

Stable rewards show that the agent is following a more consistent policy. It is also evidence that the learned strategy can perform reasonably well even when there is uncertainty in the environment.

Although small reward variations may continue because the environment is stochastic, a stable overall trend indicates improved learning, policy quality, and robustness.

Conclusion:

The early fluctuations represent exploration and learning, while the stability after 500 episodes indicates that the agent has developed a reliable policy and is moving toward convergence.